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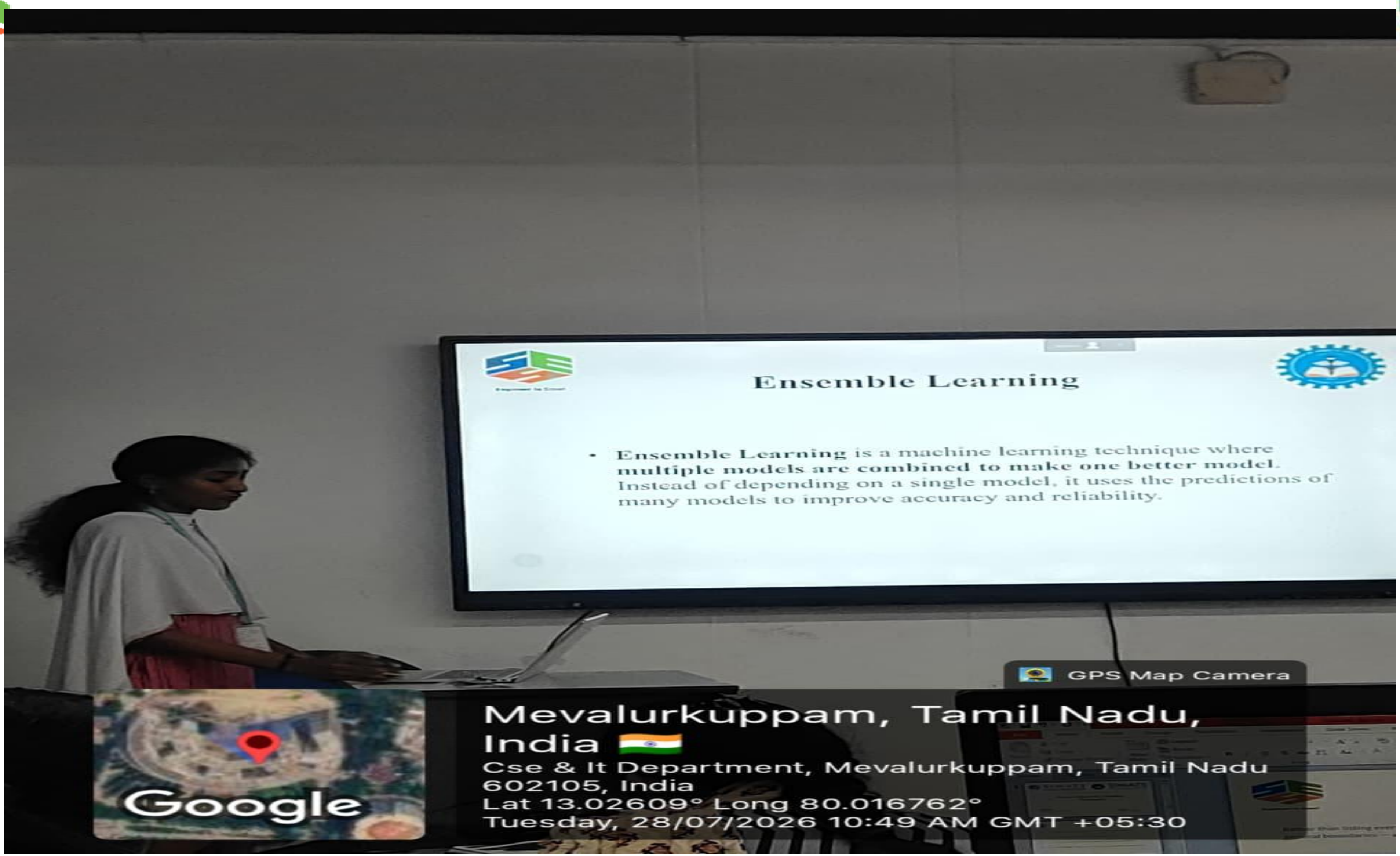

“Ensemble Learning”

COURSE CODE:ITA0608

COURSE NAME: MACHINE LEARNING

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Ensemble Learning

- **Ensemble Learning** is a machine learning technique where **multiple models are combined to make one better model**. Instead of depending on a single model, it uses the predictions of many models to improve accuracy and reliability.

Why is Ensemble Learning Important?

1. Improves prediction accuracy.
2. Reduces errors.
3. Makes the model more stable and reliable.
4. Helps reduce **variance** (overfitting) and **bias** (underfitting).

Main Techniques

1. Bagging

- Trains many models on different random parts of the dataset.
- Combines their predictions by voting or averaging.
- **Example:** Random Forest.

2. Boosting

- Trains models one after another.
- Each new model focuses on correcting the mistakes of the previous one.
- **Example:** AdaBoost.

3. Stacking

- Combines different types of models.
- A final model, called a **meta-model**, learns from their outputs and gives the final prediction.

Popular Algorithms

- **Random Forest:** Uses many decision trees with bagging to improve accuracy and reduce overfitting.
- **AdaBoost:** Uses boosting by giving more attention to wrongly predicted data, improving overall performance.

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Conclusion

Ensemble learning combines the strengths of multiple models. This improves accuracy, reduces variance and bias, and gives better predictions than using a single model.

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Thank you